
Ordered sets

Categories with particularly tidy arrow structures.

In the category of natural numbers we just saw, everything sits neatly in a straight line. This is another special kind of category, corresponding to a mathematical structure that is studied by other mathematicians in its own right, called a totally ordered set.

10.1 Totally ordered sets

An ordered set is a set equipped with an ordering. An ordering is a way of organizing all the objects in an order. A *totally* ordered set is one that is “totally organized”. However, in order to study these things mathematically we need to say it in a more rigorous way than that. This definition takes the *visual* idea of a category being “all in a straight line” and identifies an *algebraic* property that corresponds to it.

Definition 10.1 A *totally ordered set* is a category in which for all objects a, b there is exactly one arrow between them.

That is, given any objects a and b there is an arrow $a \rightarrow b$ or there is an arrow $b \rightarrow a$ but not both, unless $a = b$ in which case the identity arrow is both of these things.

Things To Think About

T 10.1 Try checking that this is true of the category of natural numbers with arrows $a \rightarrow b$ given by $a \leq b$.

Given natural numbers a, b , it is definitely true that $a \leq b$ or $b \leq a$, and they can’t both be true unless $a = b$. So this category is a totally ordered set. In fact, as $\leq$ is a motivating or quintessential example of this kind of ordering, we often write an ordering as $\leq$ regardless of what kind of relation it really was.

In the normal progression of learning mathematics, you encounter ordered sets before you encounter categories. In that case the definition has to be more direct and does not go via a category. The definition looks a bit like this.

Non-categorical definition of ordered set

An *ordered set* is a set S equipped with a binary relation $\leq$ satisfying

- reflexivity: $a \leq a$ for all $a \in S$
- antisymmetry: if $a \leq b$ and $b \leq a$ then $a = b$
- transitivity: if $a \leq b$ and $b \leq c$ then $a \leq c$
- trichotomy[†]: for all $a, b \in S$, $a \leq b$ or $b \leq a$.

Note that in math we typically use the “inclusive or” which really means “and/or”. So “ $a \leq b$ or $b \leq a$ ” means “ $a \leq b$ or $b \leq a$ or both”. If we mean one of them but *not* both, that’s called the “exclusive or”. There are various reasons for preferring the inclusive or, but one is that it comes up more often and so it’s more efficient to take it as the default.

Things To Think About

- T 10.2** 1. In fact, the reflexivity condition above is redundant. Can you see how to deduce it from the other conditions?
2. In the categorical definition we appear only to have specified one condition. What happened to the other conditions?

Reflexivity is a special case of trichotomy: if we put $a = b$ in the definition of trichotomy we get “ $a \leq a$ or $a \leq a$ ”, which is reflexivity.[‡] It’s useful to state it as a condition though, because when we relax the conditions later we *will* need to specify it.

In the categorical definition we don’t need the other conditions because:

- Reflexivity comes from the category having identities.
- Transitivity comes from composition in the category.
- Antisymmetry and trichotomy both come from there being exactly one arrow between a and b : antisymmetry comes from there being *at most* one, and trichotomy from there being *at least* one.

In normal life we don’t often think of “exactly one” to mean “at least one and at most one” but in math this is often how it pans out, as in the above example. Sometimes “at least one” is stated as “existence”, and “at most one” is “uniqueness”, but sometimes we prove that there is at most one of something before proving that there is at least one, and it’s a bit odd to think of proving that something is unique before proving it exists.

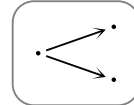
[†] “Trichotomy” ought to mean there are three cases where this looks like there are only two; however deep down it is coming from the three cases $a < b$ or $a = b$ or $a > b$.

[‡] If we say “ X is true or X is true” then it’s logically equivalent to saying “ X is true”, although it is linguistically convoluted.

Things To Think About

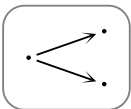
T 10.3 How many ways can you think of for a category to *fail* to be an ordered set? You could try doing this by algebra or by pictures. By algebra you'd say: what ways are there for a category to fail to fulfill the definition? By pictures you'd say: what pictures can I draw of a category that isn't just a bunch of arrows in a straight line? You don't need to say what the objects and arrows represent — it can just be a picture of dots and arrows.

Here's one to get you going. Can you see in what way this is failing to be an ordered set? I haven't labeled the objects or arrows but I'm taking them all to be different.

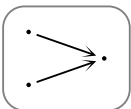


It can sometimes seem that math is all about getting right and wrong answers, but often in higher math if something doesn't work it's really interesting to try and classify all the possible ways in which it could fail. This often gives us some deeper understanding and points us in the direction of a helpful generalization in which we relax one criterion slightly in order to count some of those failures as an interesting structure in their own right.

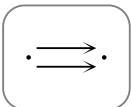
Here are some pictures of categories that are not arrows all in a line (even if we could draw them all in a physical line they wouldn't point the same way, so wouldn't be abstractly "arrows all in a line"). I haven't labeled the objects or the arrows but I'm taking them all to be different.



This one fails because it has two arrows branching out from the same place, so the arrows aren't going along in order. Formally, the definition fails as the two objects on the right have no arrow between them.



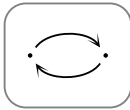
This one fails for "the same" reason as the one above, just with the branching going in rather than out. In fact we've just turned around the arrows, which is an example of what is called the dual in category theory.



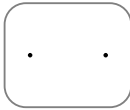
Here we have parallel arrows rather than a single straight line. This fails the definition as these two objects have two arrows between them instead of just one.



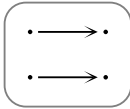
In this case we have multiple arrows even though there's only one object. They're still technically "parallel" arrows, although they're really loops.



Here's a way to make a loop via two objects. We could do this with as many objects as we like. These two objects have an arrow going both ways but are not equal.



The loop examples fail by having too many morphisms, but this one fails by having too few. The two objects have no arrows between them, not “exactly one” as in the definition.



Any disconnected category will fail, like this one. “Disconnected” means there are separate parts with a complete gap between them (but is stated more rigorously).

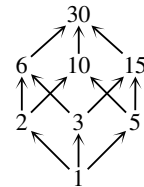
10.2 Partially ordered sets

We just saw that a totally ordered set has “exactly one” arrow between any pair of objects, and thus there are two ways to fail:

- a pair of objects with *more* than one arrow between them, or
- a pair of objects with *fewer* than one arrow between them (i.e. none).

We could now try relaxing this definition by maintaining the first condition but dropping the second, so we are thinking about categories in which there is *at most* one arrow between any two objects, but possibly none.

We have seen some of these already, such as any of the categories of factors from Chapter 5. Here's the one for 30 again:



Things To Think About

T 10.4 It might look like some pairs of objects have more than one arrow between them. For example the two composites on the right. Can you see why there is actually only one arrow there?

$$2 \longrightarrow 6 \longrightarrow 30$$

$$2 \longrightarrow 10 \longrightarrow 30$$

These two composites are equal, because an arrow $a \longrightarrow b$ is *the assertion* that a is a factor of b . Put another way, as this is a true/false relationship, there is at most one arrow $a \longrightarrow b$, so those two composites must be the same.

This is part of a more general picture: whenever there is at most one arrow between any two objects in a category then all diagrams in the category must

commute. This is because if there can only be one arrow between two objects, all composites with those endpoints must be equal.

In the example of factors there are some objects that have no arrows between them, such as 6, 10 and 15. As a result the picture is no longer a straight line but is still quite organized. It consists of several straight line paths that join up and separate in places, and no loops. These structures are interesting enough to have a name.

Definition 10.2 A *partially ordered set* is a category in which for any objects a, b there is at most one arrow between them.

For emphasis, the previous, more rigid version is called a *totally* ordered set. And these are sometimes abbreviated to “toset” and “poset” because mathematicians are so lazy.

As with tosets, you might usually meet posets before categories, via a definition that doesn’t mention categories. Just as the categorical definition of poset came from relaxing the categorical definition of toset, we could do that to the non-categorical definition as well.

Things To Think About

T 10.5 Look back at the non-categorical definition of toset and see what condition(s) we need to drop to make a non-categorical definition of poset.

The condition we need to drop is trichotomy, as we now want to allow pairs of objects that are not related either way round. Note that unlike with tosets we now can’t drop the reflexivity condition, because that was only redundant in the presence of trichotomy. This is one reason to prefer keeping it in the toset definition, so that we can then get to the poset definition by simply dropping one condition, which is more mathematically satisfying when you’re making a generalization. (Of course, I would say that this is a reason to prefer the categorical definition.)

So here’s the non-categorical definition.

Non-categorical definition of partially ordered set

A *partially ordered set* or *poset* is a set S equipped with a relation $\leq$ satisfying

- reflexivity: $a \leq a$ for all $a \in S$
- antisymmetry: if $a \leq b$ and $b \leq a$ then $a = b$
- transitivity: if $a \leq b$ and $b \leq c$ then $a \leq c$.

Things To Think About

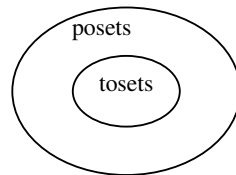
T 10.6 Look back at our pictures of categories that were not tosets. Which ones are actually posets?

The ones that are posets will be the ones that failed to be tosets only by having too *few* arrows somewhere, not too many. So not the ones with parallel arrows or loops.

This way of generalizing a definition by relaxing a condition is typical in math. We took a definition with several conditions on a structure, say conditions A, B, C, D. Then we relaxed the definition by dropping condition D and looking at structures that only satisfy A, B, C. Logically, anything that satisfies A, B, C, D definitely satisfies A, B, C and so anything that is the first type of structure will definitely be the second type, which is why the second type is called a generalization; conversely the first type is a special case.

Then it's good to ask whether there actually are any structures of the second type that aren't the first type. We have just seen some small examples of posets that are not tosets, so we know that this is a genuine generalization.

We could also encapsulate this in a Venn* diagram, as shown, or we could use logical terms and say “ $\text{toset} \Rightarrow \text{poset}$ ” although this isn't quite grammatical because “toset” and “poset” aren't statements. However, I'm not pedantic enough to make them into full statements at the expense of concision and clarity.[†]



Things To Think About

T 10.7 In Chapter 5 we moved from lattices of factors to lattices of subsets. Can you make a formal definition of a category of subsets of S (for some fixed set S , based on Section 5.6? Can you then show that it is a poset?

We start by fixing a set S . We can then construct a category with

- objects: all the subsets of S
- arrows: given subsets A, B , we have an arrow $A \rightarrow B$ whenever A is a subset of B , which we write as $A \subseteq B$.

[†] In *The Art of Logic* I characterized pedantry as precision that does not increase clarity.

Everything else follows more or less as for factors. The set of all subsets of S is called the *power set* of S , and is often written $\mathcal{P}(S)$.

Does it seem confusing that the objects of this category are now sets? It's worth just thinking about that for a second. We are going to see that one of the powerful aspects of category theory is that we can zoom in and out and use it at different levels. The objects of a category can be points, sets, sets of sets, whole categories, entire mathematical worlds. . . And I do like the fact that many types of historically pre-existing mathematical structure can be encapsulated as a special type of category.

Things To Think About

T 10.8 Look back at Chapter 5. Can you see any other posets that we looked at in that chapter?

The diagram of quadrilaterals and their generalizations was another poset. In the next chapter we're going to look at some other examples of mathematical structures that arise naturally in their own right, like posets, but aren't as tidily organized.